

CHACALLUTA, Chile

WMO: 854060

Lat: 18.3514S

Lon: 70.3358W

Elev: 55

StdP: 100.67

Time Zone: -4.00 (CHL)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.0	13.1	7.8	6.6	13.1	8.3	6.8	13.8	8.0	18.4	7.3	17.8	2.5	80	0.328

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	5.3	26.8	20.7	26.0	19.9	25.2	19.5	21.9	25.6	20.9	24.8	20.2	24.3	5.7	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.3	15.1	24.9	19.2	14.1	23.9	18.2	13.2	23.2	64.3	25.5	61.0	24.7	58.0	24.5	23.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.1	7.3	6.7	DB	9.7	28.5	1.3	1.3	8.8	29.4	8.0	30.2	7.2	30.9	6.3	31.8	(4)
(5)			WB	8.0	21.6	1.1	1.2	7.2	22.5	6.6	23.1	6.0	23.8	5.2	24.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	19.2	22.7	23.0	22.1	20.1	18.3	17.0	16.1	16.2	16.8	18.0	19.4	21.1
	DBStd	2.73	1.29	1.19	1.27	1.41	1.58	1.31	1.34	1.21	1.17	1.17	1.11	1.21
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	273	0	0	0	2	20	43	69	67	48	20	3	0
	CDD10.0	3364	392	364	376	302	258	209	190	193	205	248	283	343
	CDD18.3	594	134	131	117	54	19	3	1	2	2	10	37	84
	CDH23.3	1103	348	396	255	37	5	0	0	2	0	0	1	59
	CDH26.7	24	8	11	4	0	1	0	0	1	0	0	0	0
Wind	WSAvg	2.8	3.1	3.0	2.9	2.7	2.5	2.4	2.4	2.6	2.9	3.1	3.2	3.1
Precipitation	PrecAvg	7	1	0	1	0	0	0	1	1	1	0	0	0
	PrecMax	20	7	2	3	1	3	3	10	5	8	3	1	3
	PrecMin	1	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	4	2	0	1	0	1	1	2	1	2	1	0	1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.8	28.0	27.2	25.2	23.9	21.8	20.8	21.2	21.1	22.1	23.2	25.2
		MCWB	21.9	21.5	20.7	19.3	18.9	17.1	16.4	17.6	17.4	17.4	17.5	19.3
	2%	DB	26.2	26.9	26.1	24.2	22.5	20.2	19.2	19.2	19.9	21.1	22.8	24.6
		MCWB	20.4	20.5	20.0	18.6	17.6	15.9	15.5	16.0	16.3	16.5	17.3	18.9
	5%	DB	25.8	26.1	25.2	23.7	21.8	19.8	18.8	18.2	19.1	20.2	22.1	24.0
		MCWB	20.0	19.8	19.4	18.2	17.2	15.6	15.2	14.5	15.5	16.0	16.8	18.3
	10%	DB	25.1	25.2	24.9	22.9	21.0	19.0	18.0	17.8	18.3	19.9	21.5	23.2
		MCWB	19.6	19.5	19.2	17.8	16.7	15.2	14.4	14.1	14.7	15.8	16.6	17.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.1	22.5	21.9	20.1	19.5	17.9	17.1	18.3	18.0	18.7	18.7	20.9
		MCDB	26.1	26.1	25.7	24.0	22.8	20.2	19.4	20.9	20.1	20.5	21.8	23.9
	2%	WB	22.2	21.6	20.9	19.3	18.3	16.6	16.2	16.5	16.9	17.6	18.1	19.9
		MCDB	25.6	25.6	24.9	23.4	21.9	19.5	18.8	19.1	19.3	19.8	21.6	23.3
	5%	WB	21.1	20.8	20.2	18.6	17.5	15.9	15.5	15.1	16.0	16.7	17.5	19.1
		MCDB	24.8	25.0	24.3	22.7	21.0	18.8	18.1	17.5	18.6	19.5	21.2	23.1
	10%	WB	19.9	20.1	19.6	18.1	16.9	15.5	14.7	14.4	15.1	16.0	17.0	18.5
		MCDB	24.0	24.5	23.9	22.2	20.3	18.5	17.4	17.0	17.8	19.4	20.8	22.6
Mean Daily Temperature Range	5% DB	MDBR	5.2	5.3	5.6	5.6	5.1	3.9	3.4	3.1	3.3	3.7	4.4	4.9
		MCDBR	5.2	5.4	5.7	6.0	5.8	4.7	3.8	3.4	3.7	4.0	4.7	4.9
		MCWBR	2.7	2.8	3.1	3.6	3.5	2.9	2.3	2.0	2.1	2.2	2.6	2.7
	5% WB	MCDBR	4.4	4.8	5.4	5.5	5.7	4.4	3.5	3.3	3.6	3.9	4.5	4.7
		MCWBR	2.4	2.7	3.1	3.4	3.7	2.9	2.5	2.2	2.2	2.4	2.7	2.8
Clear-Sky Solar Irradiance	taub	0.515	0.521	0.505	0.452	0.404	0.385	0.386	0.400	0.417	0.428	0.432	0.483	
	taud	2.069	2.055	2.094	2.199	2.274	2.327	2.323	2.267	2.219	2.201	2.218	2.110	
	Ebn at Noon	843	827	817	822	827	826	837	858	881	898	909	869	
	Edh at Noon	178	178	166	139	120	110	113	128	144	152	152	171	
All-Sky Solar Radiation	RadAvg	7.53	7.18	6.90	5.78	4.63	3.75	3.76	4.55	5.60	6.51	7.53	7.81	
	RadStd	0.42	0.49	0.34	0.45	0.24	0.25	0.27	0.30	0.36	0.29	0.34	0.35	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page